

Biologically Plausible Schema Learning

What makes this room a classroom

.... What *IS* a classroom?..... What *is* a kitchen...

Goal

Schemas, in cognitive science, refer to structured sets of knowledge about common spaces or environments, that include typical objects and their arrangements within those spaces.

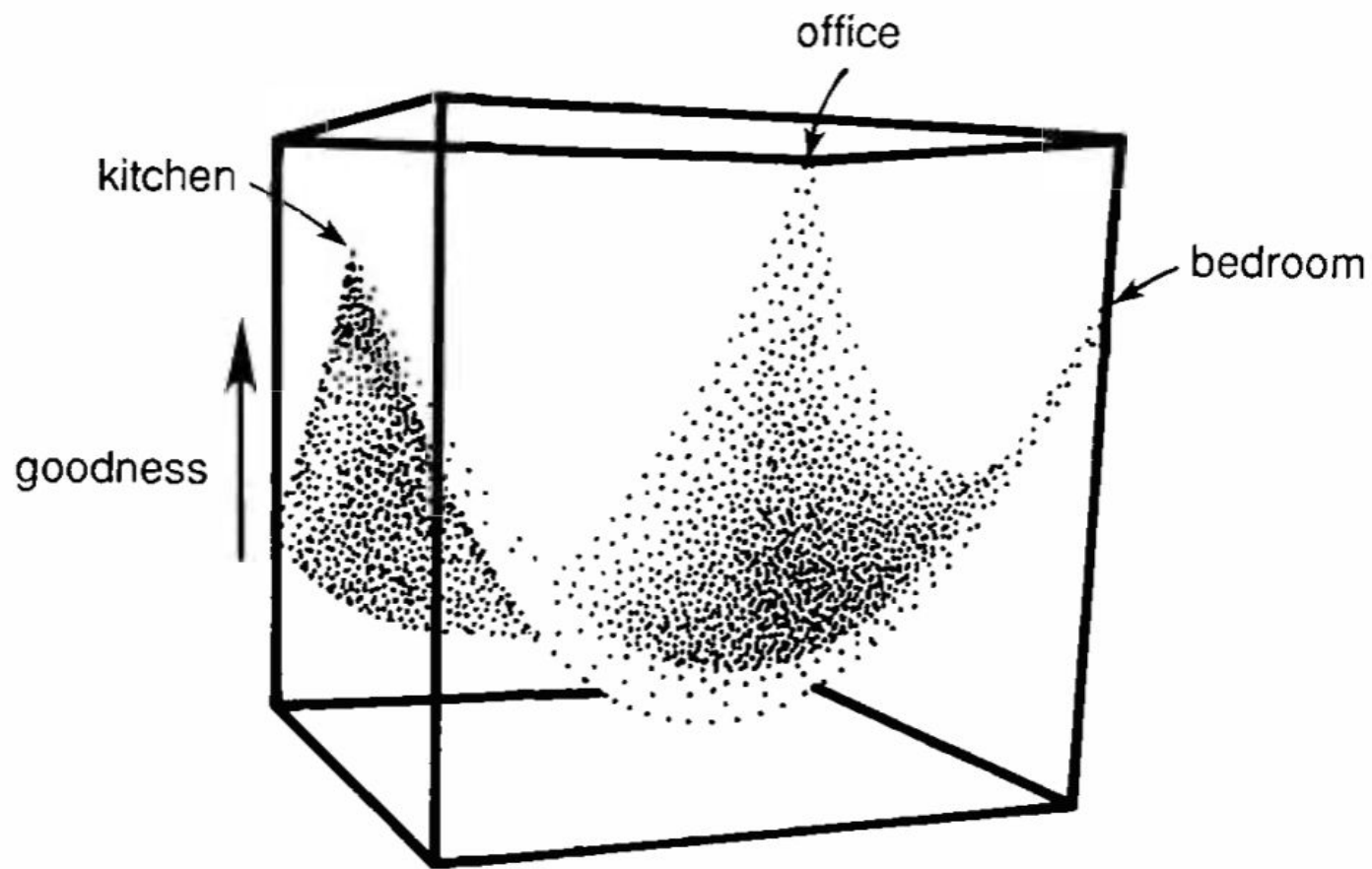
How can we model this at a biologically realistic level? Spatial learning and memory are important for navigation and formation of episodic memories. The hippocampus and medial entorhinal cortex are key brain areas for spatial learning and memory.

Motivation

TABLE 1

THE FORTY ROOM DESCRIPTORS

ceiling	walls	door	windows	very-large
large	medium	small	very-small	desk
telephone	bed	typewriter	bookshelf	carpet
books	desk-chair	clock	picture	floor-lamp
sofa	easy-chair	coffee-cup	ashtray	fireplace
drapes	stove	coffeepot	refrigerator	toaster
cupboard	sink	dresser	television	bathtub
toilet	scale	oven	computer	clothes-hanger

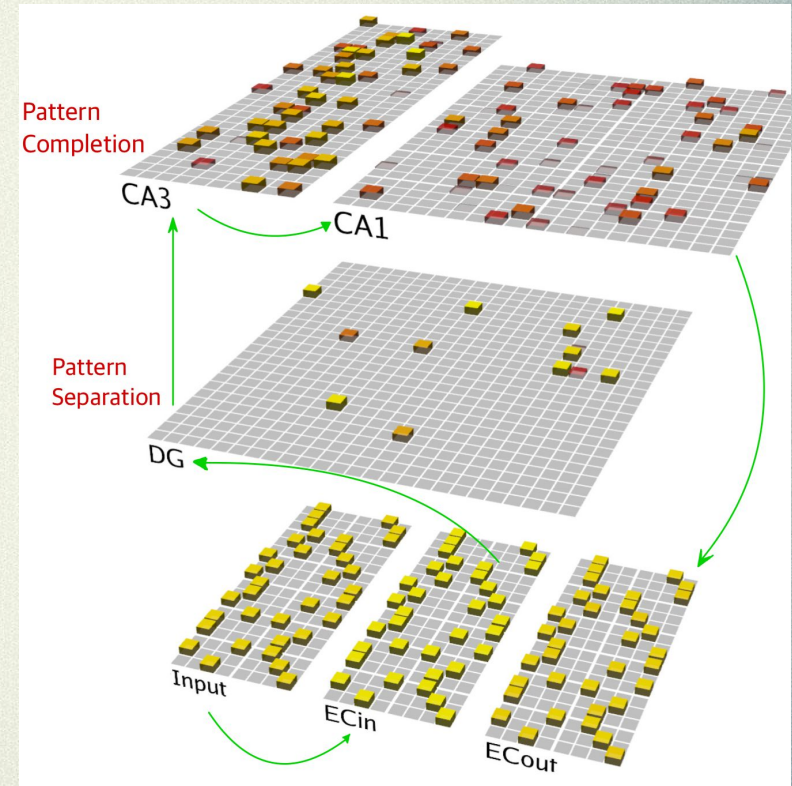


Approach : Paired Associate Learning & Hippocampus model

In the hippocampus paired associate learning involves the encoding, storage, and retrieval of these associations

AB/AC list learning, each stimulus: an item from list A, a corresponding item from either list B or C, and a "context" vector.

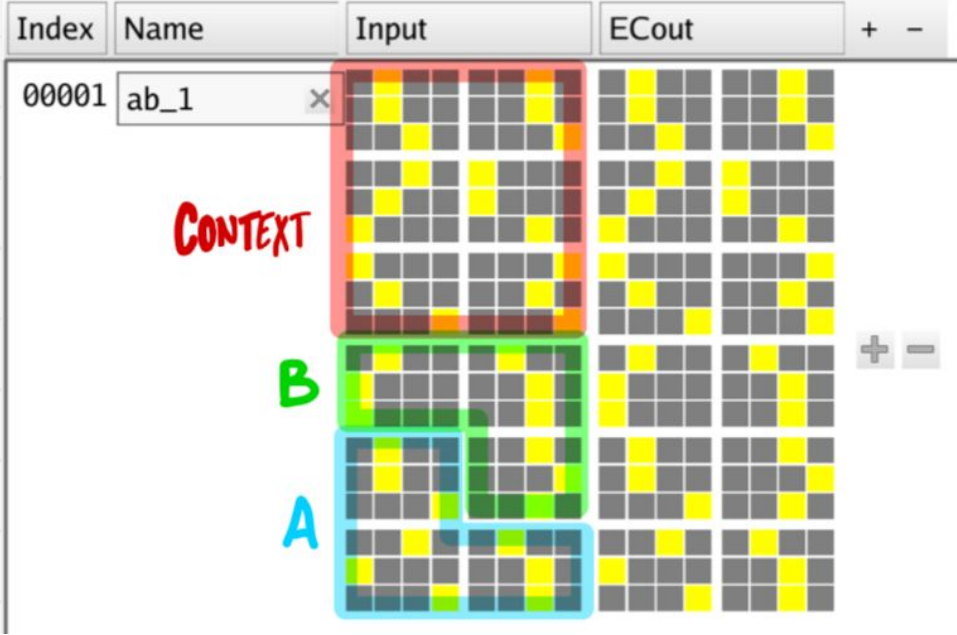
This context vector serves to differentiate whether the association is of the AB or AC type



Approach: Patterns

Window	Large	Medium	Small
very-small	coffee-pot	bathtub	bed
type-writer	book-shelf	toilet	sink

fridge	coat-hanger	cup-board	desk
sofa	picture	carpet	scale
toaster	tele-phone	computer	oven



desk-chair	clock	T.V	floor-lamp
books	easy-chair	coffee-cup	ash-tray
fire-place	drapes	stove	dresser

Take Home Message

What does this mean for how we encode room schemas with pattern separation attractor dynamics?

How different input vector encodings affect interference between the two houses.

Does the same attractor structure hold up in a biologically grounded model?

If many of the rooms are identical across the two houses, does this cause problems for the rooms that are not identical?

Questions?